

# Automated and Explainable Outcome Prediction in NSCLC: Beyond Manual Feature Extraction using LLM Embeddings

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## INTRODUCTION

- **Clinical notes** contain rich, unstructured **prognostic information**.
- **Manual extraction** is labor-intensive, slow, and **not scalable** for large cohorts.
- **LLM embeddings** can automate this process by generating **context-aware features**.
- **Explainability** is essential to translate AI predictions into **trustworthy clinical insights**.

## GOALS

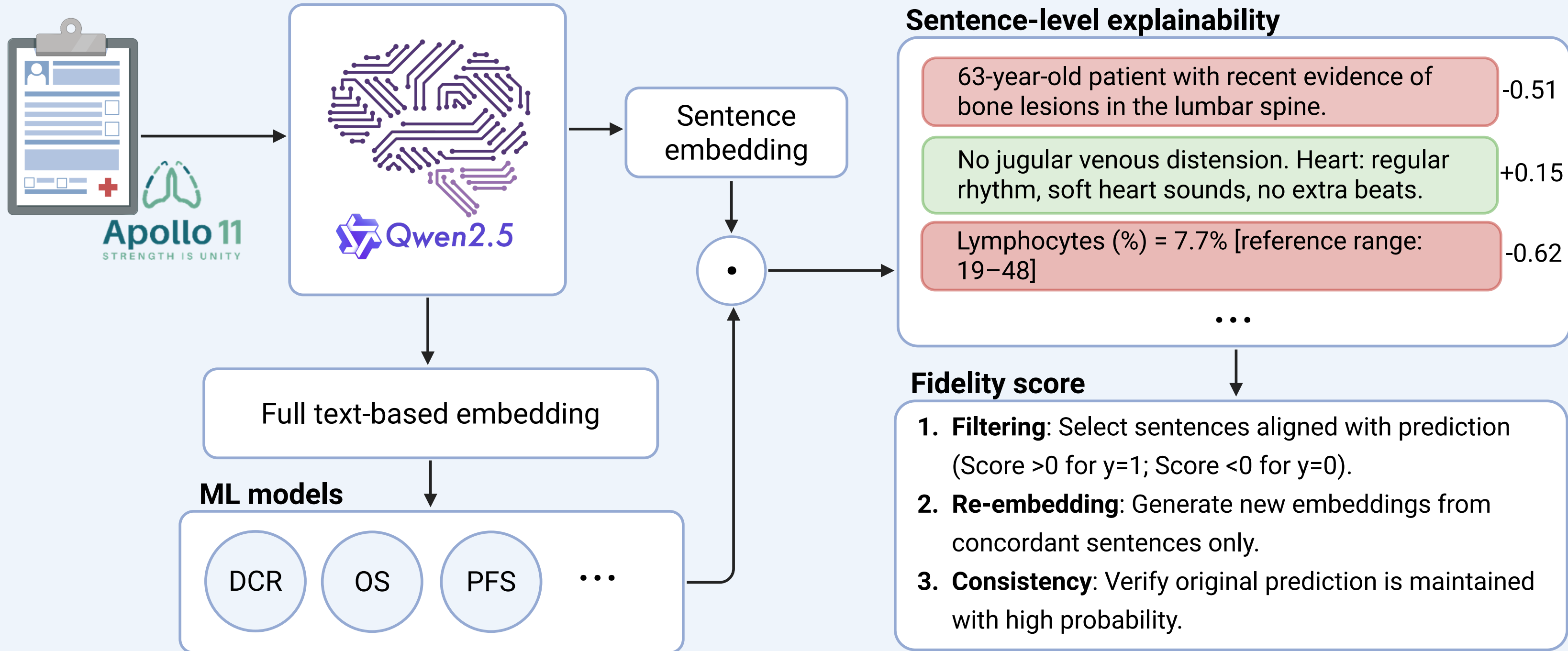
- Automate **feature engineering** from **unstructured notes**.
- Predict **OS, PFS, DCR**, and **toxicity** across **IO, CHT, and IO +CHT**.
- Provide **sentence-level explainability** for clinical transparency.
- **Validate** AI logic via a rigorous **Fidelity framework**.

## DATASET

- **691 NSCLC patients**
  - Istituto Nazionale dei Tumori
- **Apollo 11** project
  - NCT05550961
- **Heterogeneous cohort** including patients treated with
  - **Immunotherapy**
  - **Chemotherapy**
  - **Immuno-Chemotherapy**
- Multimodal analysis of
  - unstructured **clinical reports**
  - **laboratory results**
  - **genomic profiling**

## METHODS

- **Qwen2.5-7B** generates patient-level embeddings
- Implemented on a proprietary **on-premise server** to ensure data privacy and **GDPR compliance**
- **Embeddings** → **prediction models**
- Sentence-level **explainability** + **fidelity**



## RESULTS

- Accurate prediction of **OS, PFS, DCR, toxicity**
- Explainability → key **clinical drivers**
- **Fidelity** validated explanations
- Fully **automated pipeline**

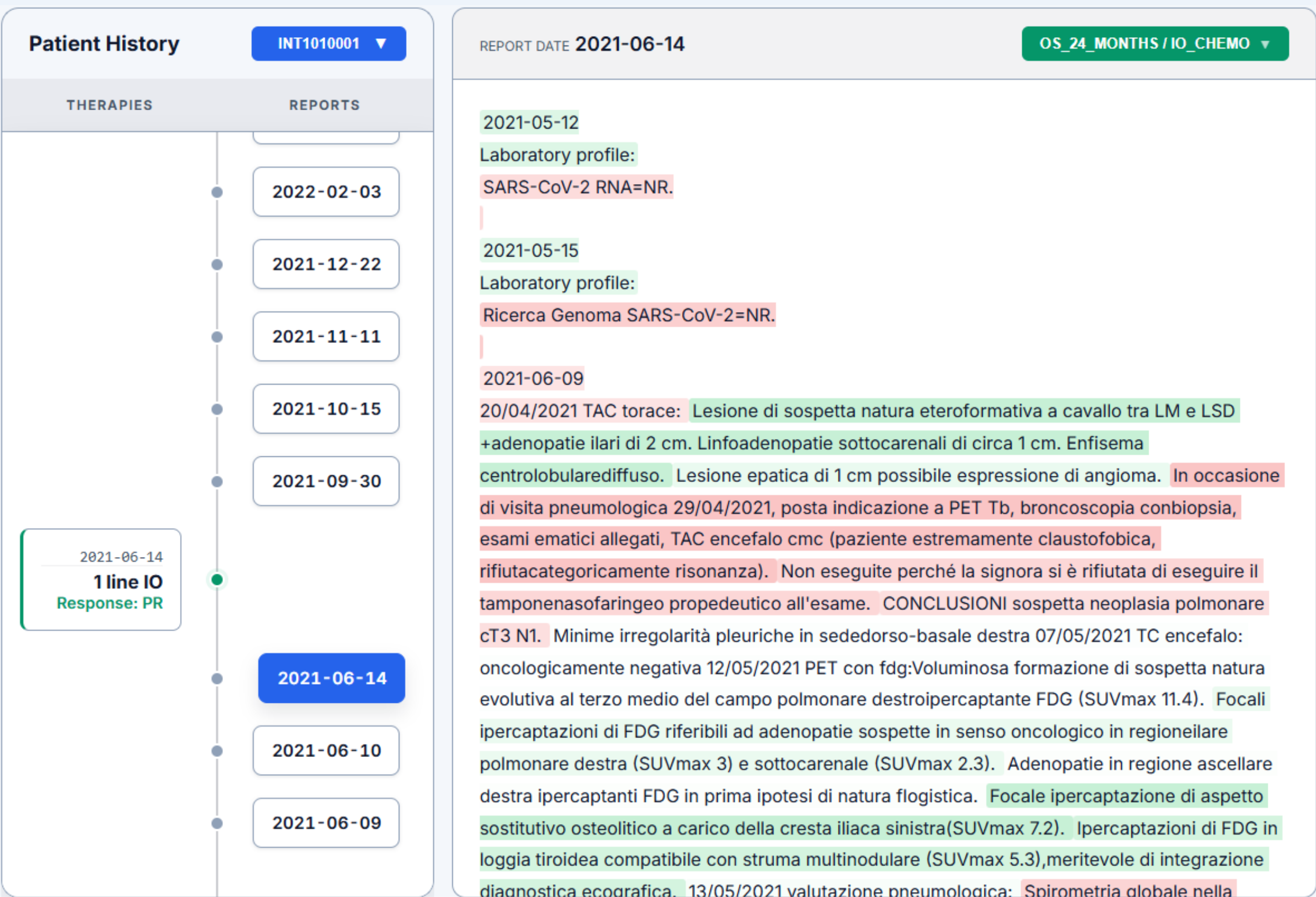
| Outcome  | IO          | CHT         | IO+CHT      |
|----------|-------------|-------------|-------------|
| OS 6     | 0.63 ± 0.13 | 0.79 ± 0.19 | 0.72 ± 0.20 |
| OS 24    | 0.67 ± 0.16 | 0.81 ± 0.19 | 0.62 ± 0.22 |
| DCR      | 0.55 ± 0.14 | 0.73 ± 0.19 | 0.56 ± 0.21 |
| TOXICITY | 0.57 ± 0.15 | *           | 0.57 ± 0.24 |

Table 1: 5-fold CV AUC (Mean ± CI)

| Outcome | IO          | CHT         | IO+CHT      |
|---------|-------------|-------------|-------------|
| OS      | 0.58 ± 0.04 | 0.68 ± 0.06 | 0.62 ± 0.06 |
| PFS     | 0.61 ± 0.08 | *           | 0.85 ± 0.10 |

Table 2: 5-fold CV C-index (Mean ± CI)

\* Not assessed due to insufficient sample size for robust training.



Sentence-level explainability (OS 24, IO+CHT arm): green highlights favorable clinical factors, red indicates unfavorable ones.

## CONCLUSION

- **LLM embeddings** replace manual feature engineering.
- **Transparent and reliable** clinical predictions.
- **Scalable** for real-world oncology.

### Next Steps:

- **Longitudinal Analysis:** Capture disease evolution
- **Digital Twin:** Dynamic patient simulation



Project details & contacts

